

Monitoring the Water Extents of the Tiber River Flowing North of the Rome

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SCRIPT: <https://code.earthengine.google.com/6730ed68e222665226e102c916760485>

1. Introduction

Climate change and urbanization are putting increasing stress on water systems, which require increased significance on river monitoring than ever before. Italy's third longest river and lifeblood of Rome, the Tiber River, requires intense monitoring because changing conditions affect its flow patterns.

This study monitors Tiber River water levels north of Rome using Landsat satellite observations between spring and summer months of 2022-2023. I want to observe March-August to document the transition from wet winter to dry summer - when Mediterranean rivers are most likely to show their greatest changes. I use different water detection indices (NDWI, MNDWI, and AWEI) in our methodology to accurately identify water surfaces across the 1,851,13 km² area north of Rome. These methods help distinguish actual water from shadows and urban buildings that would confuse the analysis. Through statistical processing of seasonal trends and comparison, this research attempts to illuminate Tiber River processes while establishing the value of satellite observation in water resource management.

2. Code Implementation

The Google Earth Engine deployment follows a five-step procedure aimed at ensuring comprehensive water extent observation and sound comparative study between the two research phases.

Step 1: Definition and Initialization of Study Area: The code begins by establishing the research area as a polygon geometry across the Tiber River region to the north of Rome. With caution, the coordinates were selected to cover the main river channel and associated water bodies while remaining

computationally effective. The study area spans approximately 1,851.13 km², calculated dynamically using GEE's `area()` function.

Step 2: Landsat Data Collection and Filtering: A crucial design decision was made to merge Landsat 8 and Landsat 9 collections to maximize coverage and image availability. The `loadLandsatData()` function in the code implements several criterias:

- Temporal filtering: Spring-summer periods (March-August) for both 2022 and 2023
- Spatial filtering: Images intersecting the study area
- Cloud filtering: Maximum 20% cloud cover threshold to ensure data quality

The choice of 20% cloud cover threshold represents a balance between data availability and quality. A threshold would reduce the number of available images, affecting temporal analysis robustness.

Step 3: Image Preprocessing: The preprocessing addresses two critical parts the of Landsat Level-2 data:

Cloud Masking: The `maskClouds()` function utilizes the QA_PIXEL band to identify and mask clouds and cloud shadows using bit manipulation. This approach ensures that cloud-contaminated pixels do not interfere with water classification:

Radiometric Scaling: The `scaleImage()` function applies official scaling factors to convert digital numbers to surface reflectance values, essential for accurate spectral index calculations.

Step 4: Water Index Calculations: Four water indices were implemented:

NDWI (Normalized Difference Water Index): Uses the contrast between green and near-infrared bands, effective for clear water bodies.

MNDWI (Modified NDWI): SWIR for NIR, particularly effective for turbid water and urban environments where the Tiber River flows.

AWEI variants: Two versions (AWEIsh and AWEInsh) were implemented following Feyisa et al. (2014) formulations:

- AWEIsh: Optimized for water with shadow contamination
- AWEInsh: Designed for non-shadow water areas

Step 5: Water Classification and Threshold Optimization:

Multiple classification approaches were implemented to provide comparative analysis:

- MNDWI threshold: -0.1 (optimized for turbid river water)
- NDWI threshold: 0.0 (standard threshold)
- AWEI threshold: 0.0 (based on “Feyisa, G.L., Meilby, H., Fensholt, R., & Proud, S.R. (2014). Automated Water Extraction Index: A new technique for surface water mapping using Landsat imagery.”)
- Combined approach: Intersection of MNDWI and NDWI for conservative estimates
- Aggressive approach: Union of all indices for maximum water detection

The MNDWI threshold of -0.1 was chosen to explain for the turbid nature of the Tiber River, where standard thresholds can fail to detect water accurately.

3. Results and Analysis

Analysis completed the processing of 61 Landsat images in 2022 and 43 images in 2023 across the 1,851 km² study area. MNDWI-based water classification with a threshold of -0.1 was able to effectively classify water surfaces amidst the turbid conditions of the Tiber River system. Water Area Statistics and Comparative Analysis Results showed an unexpected 38.6% increase in the mean water extent from 2022 to 2023, an absolute change of 1.51 km². Such a high increase indicates improved water availability for the spring-summer period of 2023.

Water Area Statistics and Comparative Analysis

2022: Mean: 3.91 km² | Median: 2.23 km² | Range: 0-14.22 km² | Std Dev: 4.00 km²

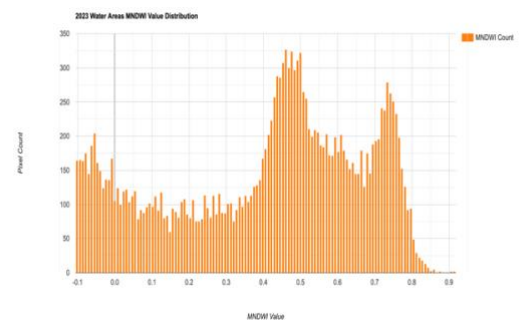
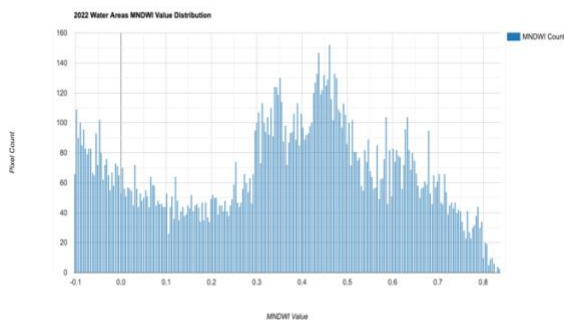
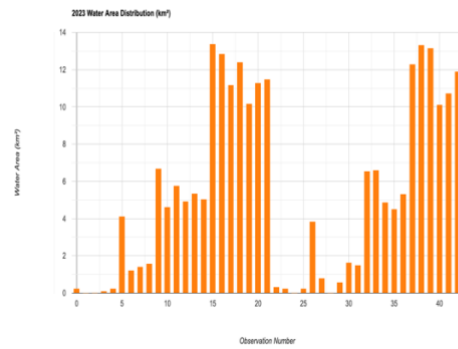
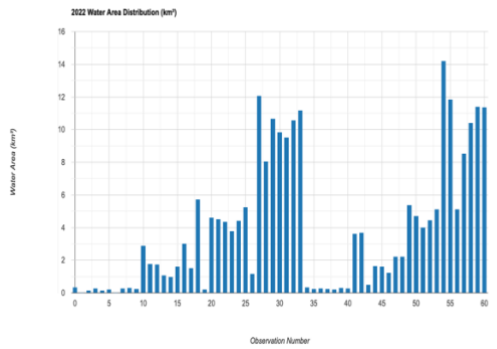
2023: Mean: 5.41 km² | Median: 4.88 km² | Range: 0-13.39 km² | Std Dev: 4.75 km²

The most significant change was for median values, with 2023 recording 118.8% more median water area (4.88 km²) compared to 2022 (2.23 km²), which indicates more stable water presence throughout the observation period.

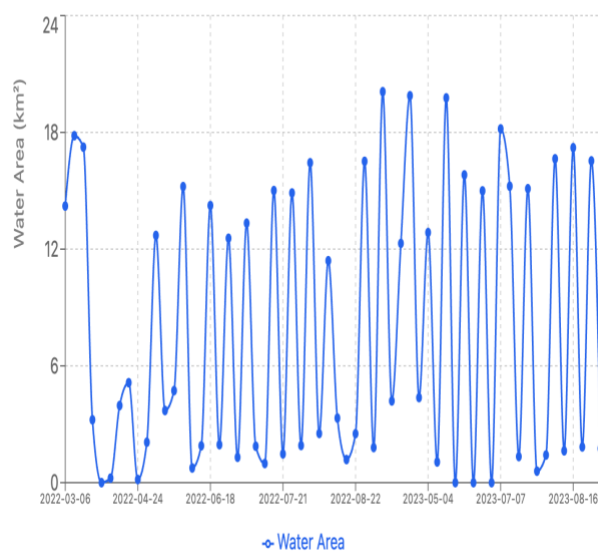
Temporal and Seasonal Patterns

Both study periods demonstrated considerable seasonal fluctuation, with water areas ranging from complete absence during extreme dry conditions to over 13-14 km² during peak availability. However, the 2023 period showed more stable water presence overall, with notably fewer instances of complete water absence compared to 2022.

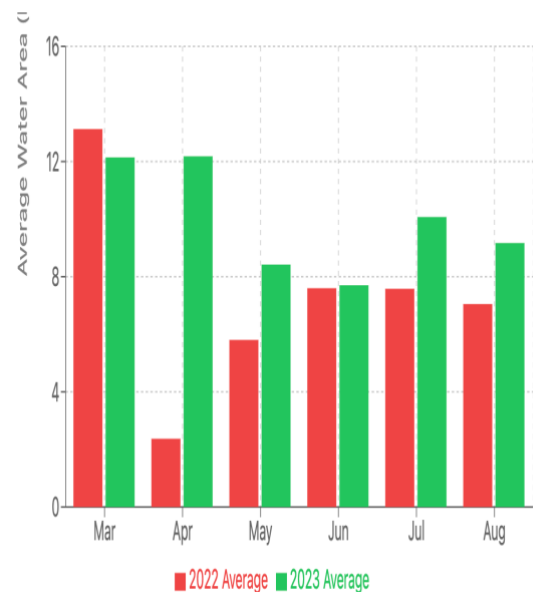
MNDWI histogram analysis confirmed that 2023 contained significantly more pixels with higher MNDWI values (>0.4), supporting the quantitative measurements of increased water area. The 2023 period displayed a broader distribution with notably higher peak values, indicating more extensive water areas with stronger spectral characteristics.



Water Area Time Series



Monthly Average Comparison



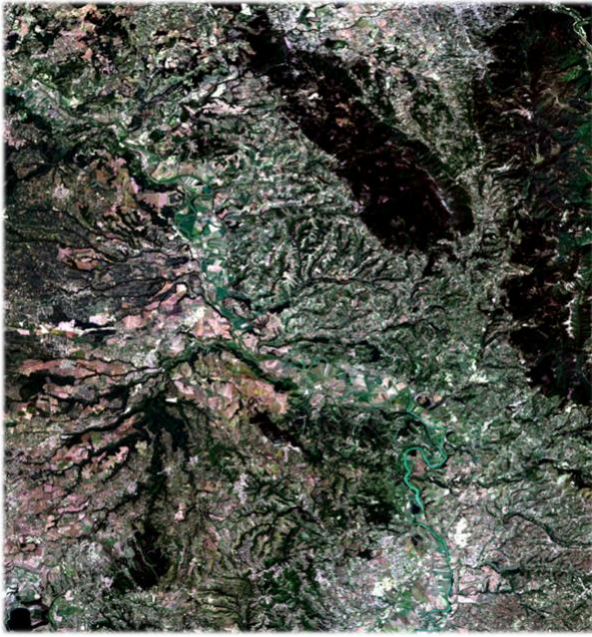


Figure 1: RGB composite of study area (2023)



Figure 2: Water Mask 2022

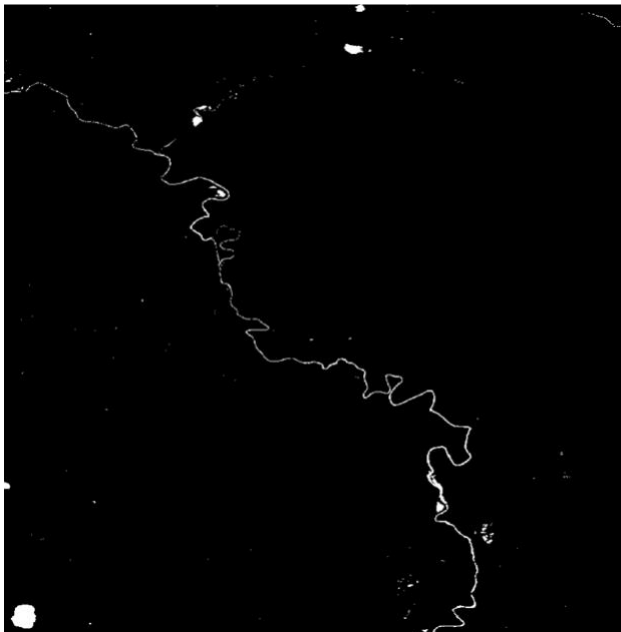


Figure 3: Water Mask 2023

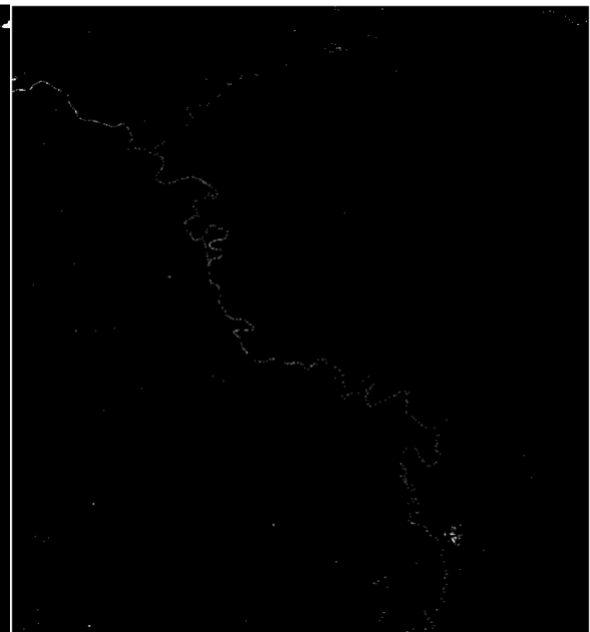


Figure 4: Water Change Between 2022-2023

4. Conclusion

This study successfully demonstrated MNDWI-based satellite monitoring for the Tiber River north of Rome, revealing a significant 38.6% increase in mean water area from 2022 to 2023 (3.91 km² to 5.41 km²). The 118.8% increase in median values indicates improved water stability during the 2023 spring-summer period. These findings establish satellite remote sensing as a valuable tool for Mediterranean river monitoring under climate change pressures.